

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

*I SARU HASINI.S (192521418)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SARU HASINI.S

Annexure A Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

A. Why does this issue occur in ECB mode?

Electronic Codebook (ECB) encrypts each plaintext block independently using the same encryption key.

- If two plaintext blocks are identical, ECB produces the **same ciphertext blocks**.
- Since the company's financial data contains **repeated data blocks**, these repetitions appear as identical patterns in the ciphertext.
- An attacker cannot directly read the data, but they can identify repeated values and infer information about the structure or content of the message, reducing confidentiality.

Example:

- Plaintext Block A → Ciphertext X
- Plaintext Block A (repeated) → Ciphertext X (same output)

This predictable behavior reveals patterns, making ECB unsuitable for sensitive or repetitive data.

B. Suggest a more secure block cipher mode and justify your choice.

A better choice is **Cipher Block Chaining (CBC) mode**.

Justification:

- In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption.
- The first block uses a random **Initialization Vector (IV)**.
- Even if two plaintext blocks are identical, their ciphertext blocks will usually be different because each block depends on the previous ciphertext.

Advantages of CBC in this scenario:

- Hides patterns in repeated financial data.
- Produces different ciphertext for identical plaintext blocks.
- Provides much stronger confidentiality than ECB when a unique, random IV is used for each message.

Conclusion:

- **ECB:** Not secure for repeated financial data because it reveals patterns in the ciphertext.
- **CBC:** More secure because it removes these patterns by chaining blocks together with a random IV, making it well-suited for transmitting sensitive financial information.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

A. Most suitable algorithm

The most suitable algorithm is **Blowfish**.

B. Justification based on security strength and efficiency

Algorithm	Security	Efficiency	Suitability
DES	Weak (56-bit key, vulnerable to brute-force attacks)	Fast	✗ Not suitable for banking systems
3-DES	More secure than DES (uses three DES operations)	Slow due to triple encryption	⚠ Secure but less efficient
Blowfish	Strong security with variable key size (32–448 bits)	Faster than 3-DES	✓ Best choice

Reasons for choosing Blowfish:

- **High security:** Supports keys up to **448 bits**, making brute-force attacks impractical.
- **Better performance:** Encrypts data faster than **3-DES** because it does not perform triple encryption.
- **Suitable for banking:** Provides a good balance between strong security and moderate performance, making it appropriate for protecting customer transactions.

Conclusion

- **Selected algorithm: Blowfish**
- **Reason:** It offers stronger security than DES and better performance than 3-DES, making it the best option among the three choices for a banking application requiring high security and moderate performance.

3. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

A. Identify the most suitable algorithm

Blowfish is the most suitable encryption algorithm for this banking system.

B. Justification based on security strength and efficiency

- **DES:** Uses a **56-bit key**, which is no longer secure because it can be broken by brute-force attacks.
- **3-DES:** Provides better security than DES by applying DES three times, but it is **slow** and less efficient for modern banking applications.
- **Blowfish:** Supports a **variable key size (32–448 bits)**, offers **strong security**, and is **faster than 3-DES**, making it a good balance between

security and performance.

Conclusion

Blowfish is the best choice among the given options because it provides **strong encryption, better efficiency than 3-DES, and much higher security than DES**, making it suitable for protecting customer banking transactions.

4. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

A. Most Suitable Algorithm

Blowfish is the most suitable encryption algorithm among **DES, 3-DES, and Blowfish**.

B. Justification

- **DES:** Uses a **56-bit key**, which is considered insecure and vulnerable to brute-force attacks.
- **3-DES:** Offers improved security over DES by encrypting data three times, but it is **slow** and less efficient for processing banking transactions.
- **Blowfish:** Supports **key sizes from 32 to 448 bits**, provides **strong security**, and is **faster than 3-DES**, making it suitable for applications requiring high security and moderate performance.

Conclusion

Blowfish is the best choice because it offers **strong encryption, better performance than 3-DES, and significantly higher security than DES**, making it appropriate for securing customer transactions in a banking system.

5. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

A. Identify the most suitable algorithm

The most suitable algorithm for this banking system is **Blowfish**.

B. Justification based on security strength and efficiency

- **DES (Data Encryption Standard):**
 - Uses a **56-bit key**, which is considered weak and can be cracked using modern brute-force techniques.
 - Not recommended for protecting sensitive banking data.
- **3-DES (Triple DES):**
 - Provides stronger security than DES by applying encryption three times.
 - However, it has **lower performance** because of the additional encryption operations, making it slower for handling large volumes of transactions.

- **Blowfish:**

- Uses a **variable key length from 32 to 448 bits**, providing strong protection against attacks.
- Has **faster encryption and better efficiency** compared to 3-DES.
- Provides a good balance between **security and performance**, which is important for banking applications.

Conclusion

Blowfish is the preferred choice because it provides **high security with efficient performance**, making it suitable for encrypting customer transactions in a banking system.

6. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

7. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

8. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand